Higher-Order but Inert: The Topology of DeFi Stablecoin Liquidity Through the USDC and Terra Depegs

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Abstract

Multi-asset automated-market-maker pools (e.g. Curve's 3pool $\{\mathrm{USDC}, \mathrm{USDT}, \mathrm{DAI}\}$) encode genuinely three-way relationships that a pairwise graph discards. We ask, using persistent homology of the *nerve complex* of pool co-membership, (i) how much of the loop structure of Ethereum stablecoin liquidity is genuinely higher-order, and (ii) whether that structure moves through systemic stress. Working entirely from the free DeFiLlama API and a level-invariant (TVL-share) filtration that neutralises the mechanical TVL drawdown, we report a split result. *Descriptively*, a majority of the independent loops in the 1-skeleton are filled by $\geq$ 3-asset simplices (52–77 % across snapshot dates), and this fraction trends down over 2022–2023—the multi-way rhetoric is right about *composition*. *Dynamically*, it is not: against a placebo-window permutation test over 632 daily complexes, the USDC-depeg window is statistically unremarkable on *every* topological *and* pairwise-graph observable (34th–82nd percentile, all $p > 0.35$), and through the Terra/UST collapse the reconstructable structure is perfectly rigid (essential $B_1 = 3$ and 1-skeleton cycle rank = 13 on all 61 days while universe TVL fell 52 %). We also document a methodological pitfall: on this object H_1 is entirely essential, so the standard persistence-landscape / L^p summary is identically zero and must be replaced by essential Betti numbers and H_0 total persistence. All findings are *survivor-conditional*: only 7.5–12.5 % of today's pools existed at the event dates, an upper bound on usable history, and this ceiling even renders one construction choice (LP-token resolution) empirically untestable. We read the null economically: stablecoins are near-substitutes pegged to a common numéraire, so their liquidity network is structurally shallow and price-stress-invariant. Code and data are fully reproducible.

1 Introduction

Classical models of financial contagion are pairwise: institutions are nodes, bilateral exposures are edges (Eisenberg & Noe, 2001). But a multi-asset AMM pool is an inherently multi-way object. Curve's 3pool binds {USDC, USDT, DAI} into a single contract; a graph records only the three pairwise projections and throws away the joint membership. Higher-order network methods—hypergraphs and simplicial complexes (Bick et al., 2023)—are designed to retain exactly this information, and a nascent literature argues they matter for systemic risk (Forte, 2026).

The decisive reason to study this in decentralised finance (DeFi) rather than traditional finance is *observability*: pool compositions and reserves are public and machine-readable on-chain, removing the confidentiality barrier that has always constrained structural contagion research. That, not the topology per se, is the strongest motivation for this work.

We make the "higher-order matters" claim testable by measurement rather than rhetoric. For each daily snapshot we build the nerve complex of pool co-membership and compute the **skeleton-vs-nerve decomposition**: the cycle rank of the 1-skeleton ($E - V + B_0$, the number of loops a graph would see) versus the Betti-1 number B_1 of the nerve (the loops that survive once multi-asset pools fill their faces). Their difference is precisely the loop structure explained by genuine higher-order co-occurrence.

Our contributions, and their honest weights, are:

1. **A structural (not return-space) topological measurement of DeFi liquidity** (Sec. 5.1). A majority of loops are higher-order, but the level is survivor-biased and only the downward *trend* is defensible.
2. **A rigorous event-study null** (Sec. 5.2). Neither topological nor pairwise-graph observables distinguish the USDC or Terra windows from placebo windows. The null is symmetric across orders and robust to thresholds, window sizes, and construction forks.
3. **A methodological caveat** (Sec. 5.3). Persistence landscapes are identically zero here; this is expected given all-essential H_1, but it is a concrete pitfall for anyone porting the Gidea–Katz return-space toolkit (Gidea & Katz, 2018) to structural complexes.
4. **An honest survivorship characterisation** (Sec. 3) severe enough to bound every claim and to render the LP-resolution design choice untestable from registry data.

A rigorous negative on (2) is itself informative: it maps where higher-order structure does *not* add value over pairwise methods, and we did not know this in advance.

2 Related work

Higher-order finance is early-stage. Forte (2026) give the first hypergraph analysis of bank–firm credit networks (Central Bank of Argentina credit registry, adjusted H-eigenvector centrality); this is TradFi credit and centrality, *not* persistent homology and *not* DeFi, and it establishes only that "higher-order beats pairwise" is a live direction. **Return-space TDA** is by contrast saturated: persistent homology of return point clouds covers equities (Gidea & Katz, 2018), FX, and cryptocurrency co-movements (FX co-movements, 2025). We keep this lineage strictly separate—our complex is built from *structure* (co-membership), not return correlation. **Graph analysis of DeFi** exists: Uniswap network (2025) model Uniswap as a token-node / pool-edge network and study centralisation and fragility. That is exactly the pairwise baseline our decomposition sits on top of, and we compute it head-to-head. To our knowledge the persistent-homology / nerve treatment of DeFi liquidity structure is previously unoccupied; TradFi/DeFi survey (2025) survey the surrounding systemic-risk landscape without it.

3 Data and the survivorship ceiling

We use two free, unauthenticated DeFiLlama endpoints: the pool registry (`yields.llama.fi/pools`: project, chain, symbol, `underlyingTokens`, current TVL, stablecoin flag) and per-pool daily TVL history (`yields.llama.fi/chart/{pool}`, reaching back to ≈February 2022). The MVP universe is Ethereum, stablecoin-flagged pools with 2–8 underlying tokens (≈600 pools).

Reconstruction and its key simplification. A pool's token set is fixed at deployment, so the historical complex at date t is today's registry filtered to pools with TVL above threshold at t. This removes subgraph/indexer dependencies from the critical path—but at a cost.

The survivorship ceiling (carry into every claim). Only **75 of ≈602 (≈12.5 %)** of today's pools were live at the USDC depeg (2023-03-11), and **45 of 599 (7.5 %)** at the Terra crash (2022-05-10). Crucially these are *upper bounds on usable history* ("how many current pools are old enough"), not true survivorship: pools delisted before today are unrecoverable from this API, and for Terra the signal-bearing pools (UST, Anchor-adjacent) are precisely the delisted ones. Every result below is survivor-conditional, and we flag where this most threatens interpretation.

4 Method

Complex. Vertices are tokens (keyed by contract address). A pool with token set σ contributes the filled simplex on σ (downward closure taken): a 2-token pool is an edge, a 3-token pool a filled triangle, and so on. This is the *nerve*, not a flag complex—observed loops are genuine coverage holes, the economically meaningful object. We track B_0 (fragmentation), essential B_1 (loops), B_2 (voids; found $\equiv 0$), the 1-skeleton cycle rank $E - V + B_0$, the higher-order *gap* $= (E - V + B_0) - B_1$, and H_0 total persistence (a merging / flight-to-coordination proxy).

Filtration (the TVL confound). Raw dollar TVL craters mechanically in a depeg because price craters, so a TVL-filtered complex would shrink for reasons of price, not structure—a laundered copy of the drawdown baseline. We therefore filter by each pool's *share* of universe TVL at date t, entering strong pools first at $f = -\log_{10}(\text{share})$. A result that dies under share weighting is a price artefact; ours are reported on share weights throughout, with raw-TVL as a checked robustness fork.

Inference (non-parametric only). With $N = 2$ events no cross-event or parametric machinery is valid. We use a **placebo-window permutation test**: slide the $[-30, +30]$-day window across every non-overlapping centre date in a 632-day baseline (2022-04 to 2023-12) and locate the event window's statistic in that placebo distribution, reporting a two-sided empirical p and percentile. Because the series is near-piecewise-constant we also report circular block-bootstrap CIs and a run-length diagnostic, and—critically—an empirical p-value floor $1/(\#\text{placebos})$ so the test cannot over-claim.

Baselines and validation. We compute seven pairwise 1-skeleton graph metrics (edges, density, components, transitivity, spectral radius, share-weighted spectral radius, degree dispersion) on the identical construction—the graph is *asserted at runtime* to be the 1-skeleton of the same complex (V, E, B_0 agree)—and put them through the same placebo test, so RQ3 is answered head-to-head. Topology code is validated on hand-checkable toy complexes (a hollow triangle has one loop; a filled one none; a tetrahedron boundary has $B_2 = 1$), and construction knobs (dust cap, window size, LP fork) are swept for robustness.

The LP-resolution fork, and why it is moot. Curve metapools nominally read $\{X, \text{3CRV}\}$; whether 3CRV is one vertex or resolved to {USDC,USDT,DAI} changes the complex. We implement both. But DeFiLlama's `underlyingTokens` already returns base assets, and **zero of ≈600 current pools embed the 3CRV triple as a strict subset**—the metapools the fork would act

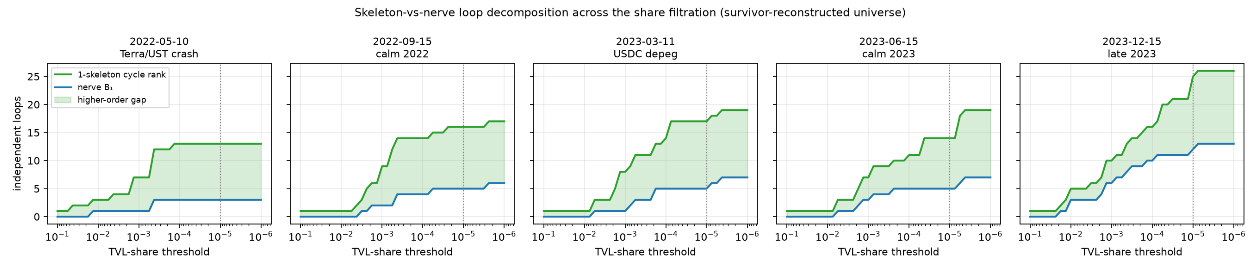

Figure 1: Skeleton-vs-nerve loop decomposition across the TVL-share filtration (deep liquidity on the left; dotted line is the operating threshold 10^{-5}). Green: 1-skeleton cycle rank (loops a graph sees). Blue: nerve B_1 (loops surviving higher-order fills). Shaded: the higher-order gap. The gap is a large, persistent share of the loop structure at every date and in both crashes, but its *shape* is essentially fixed—the central descriptive object of the paper.

on did not survive. The two forks are therefore identical on all real data; the choice is untestable from registry reconstruction, a direct casualty of survivorship.

5 Results

5.1 Descriptive: liquidity is higher-order in composition (RQ1)

Figure 1 plots the skeleton-vs-nerve decomposition across the share filtration at five snapshot dates. At the operating threshold (10^{-5}), the fraction of 1-skeleton loops filled by ≥ 3-asset pools is:

date	2022-05-10	2022-09-15	2023-03-11	2023-06-15	2023-12-15
higher-order fraction	77 %	69 %	71 %	64 %	52 %

Two claims. (i) *In the survivor-reconstructed universe, a majority of loop structure is genuinely higher-order*: the multi-way rhetoric is correct about composition. (ii) The fraction *trends down* 77 % → 52 % over 19 months (not strictly monotone), consistent with a transition from the 2022–23 Curve-metapool era toward a pair-dominated (Uniswap-era) regime. We emphasise the survivor-conditional caveat sharply here: multi-asset survivors are old, large pools, so the *level* is likely biased upward; the *trend* direction is the more defensible statement, and even it could in principle reflect differential delisting.

5.2 Dynamic: nothing moves, at any order (RQ2/RQ3)

USDC depeg. Table 1 places the event window in the placebo distribution for every observable. *The depeg window is unremarkable on all of them*: topological metrics land at the 35th–82nd percentile and pairwise metrics at the 32nd–65th, none approaching significance (all $p > 0.35$; essential B_1 at the 54th percentile, $p = 0.93$). The honest RQ3 answer is therefore *symmetric*: it is not that topology fails where graphs succeed—the share-filtered structure is event-invariant at *every* order. Block-bootstrap 95% CIs are tight (essential $B_1 \in [4.5, 5.0]$) because the series is nearly constant (essential B_1 takes 11 distinct values with a 153-day longest run over the full span), and the result is stable under thresholds (10^{-4}–10^{-6}), window sizes (30/45/60 d) and both LP forks.

metric	kind	event percentile	two-sided p
edges	graph	64.7	0.71
density	graph	31.8	0.64
components	graph	46.4	0.93
transitivity	graph	46.4	0.93
spectral radius	graph	53.6	0.93
weighted spectral radius	graph	46.6	0.93
degree dispersion	graph	53.6	0.93
essential B_1	topology	53.6	0.93
higher-order gap	topology	82.2	0.36
H_0 total persistence	topology	34.9	0.70
B_0	topology	46.4	0.93

Table 1: Placebo-window permutation test for the USDC-depeg window (2023-03-11, $[-30, +30]$ d) over 632 daily complexes. No metric of either kind is anomalous.

Terra/UST. The reconstructable structure is *perfectly rigid* through the collapse: essential $B_1 = 3$ and 1-skeleton cycle rank = 13 on all 61 days of the window while universe TVL fell 52 % (\$11.4B $\rightarrow$ \$5.5B). Only H_0 total persistence drifts mildly ($\approx 2.9 \rightarrow 2.6$ through crash week, consistent with slight merging). This is the strongest form of the rigidity result—the largest collapse in DeFi history does not move a single loop of the survivor-reconstructed complex.

A placebo-test trap we flag rather than exploit. The naive permutation test returns spuriously "significant" Terra extremes (essential B_1 at the 0th percentile; several graph metrics $p \approx 0.007$). These are artefacts: the Terra window sits at the left edge of a *non-stationary* series in which the universe grows monotonically ($45 \rightarrow 96$ pools over the span), so every size-correlated metric is mechanically extreme against later placebos. Exchangeability fails at the series edge; we therefore rely only on the within-window dynamics for Terra, which are flat. We report this because it is exactly the kind of edge-effect that produces false positives in short-series event studies.

5.3 Methodological: persistence landscapes are degenerate here

Under the share filtration, *every* H_1 class is essential: across all 579 days of an 18-month span (and in both event windows) there are **zero finite H_1 bars**. A loop that forms among strong pools is never closed by a weaker one. Consequently the standard persistence-landscape / L^p summary—on which essentially all return-space TDA-in-finance work runs—evaluates to *identically zero* in H_1 (exact $L_1 = L_2 = L_\infty = 0$, every day) under the usual convention of discarding infinite bars. Truncating infinite bars at a cutoff makes the norm nonzero but arbitrary: it shifts $\sim 10\%$ with the (unprincipled) cutoff choice and still does not separate event from placebo (L_2 mean 1.22 vs 1.25). An H_0 control (which has genuine finite bars) yields nonzero landscapes ($L_2 \approx 0.62$ vs 0.46), confirming the machinery is not broken—the degeneracy is specific to H_1 on this object.

We are candid about the weight of this: that landscapes discard essential classes is textbook, and all-essential H_1 is a fairly direct consequence of the construction. The contribution is a *practitioner's caveat*: apply the Gidea–Katz pipeline naively to a structural liquidity complex and you will read identically zero and wrongly conclude "no topology." The correct observables are essential Betti numbers and H_0 total persistence, which is what we track.

6 Discussion

The descriptive and dynamic findings resolve cleanly under one economic reading. Stablecoins are near-perfect substitutes pegged to a *common numéraire* (the US dollar). The liquidity network is thus a hub-and-spoke around "dollar-ness"—mostly copies of the 3pool and shallow wrappers—rather than a rich web of distinct multi-way dependencies. Two consequences follow directly: the multi-way *composition* is real (pools genuinely bundle three-plus dollars, hence the large gap), but the *shape is baked in and near-static*, because a depeg plays out as prices and flows moving *within* a fixed co-membership scaffold, not as the scaffold rewiring. Topology detects shape; when shape does not change, topology—at any order—has nothing to detect beyond what H_0/connectivity (already a pairwise quantity) captures.

This delimits where structural higher-order TDA might pay off: slower, deeper regime changes that actually re-wire membership (protocol migrations, sustained delistings); non-substitute asset universes (mixed collateral, cross-asset lending) where multi-way dependencies are heterogeneous; and—above all—*de-censored* data in which the pools that die during a crisis remain in the complex. None of these are reachable from the free registry endpoint alone.

7 Limitations

- **Survivorship** is pervasive and unfixable with this source: 7.5–12.5% usable-history *upper* bounds, true dead-fractions unrecoverable, and one design fork (LP resolution) rendered untestable. All claims are survivor-conditional.
- $N = 2$ **events**, both survivor-censored; no cross-event inference. The Terra placebo percentiles are invalid (edge effect) and we do not use them.
- **Single chain, stablecoins only**; the RQ1 *level* is survivor-biased upward and only the trend is defensible.
- **Construction choices** (nerve vs. flag; share filtration; 10^{-5} dust cap; 3-day forward-fill for data gaps) are defended and swept but not exhaustive.
- The **methodological caveat is expected**, not deep; we frame it as a pitfall, not a theorem.

8 Conclusion

Measured rather than asserted, the higher-order structure of Ethereum stablecoin liquidity is substantial in *composition* (a majority of loops are higher-order fills, with a declining trend) but *inert* in dynamics: neither topological nor pairwise-graph observables distinguish the USDC or Terra depegs from ordinary windows, and the structure is perfectly rigid through a 52% TVL collapse. The right summary is essential Betti numbers, not persistence landscapes, which degenerate here. The honest thesis is that a network of common-numéraire substitutes is structurally shallow and price-stress-invariant—a clean delimitation, on public and reproducible data, of where higher-order topology does and does not add value over pairwise methods, subject throughout to a severe survivorship ceiling.

Reproducibility

All code, cached observables, and figures are available at `https://github.com/adamLEVi16/topology-engine` (`defi_topology/`). The pipeline runs from the free DeFiLlama API with no authentication; toy-complex tests pin the topology, and every number above is regenerated by a single script per result. Pre-registered observable set: $\{B_0, \text{essential } B_1, \text{gap}, H_0 \text{ total persistence}\}$.

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